

The Brain Stimulator

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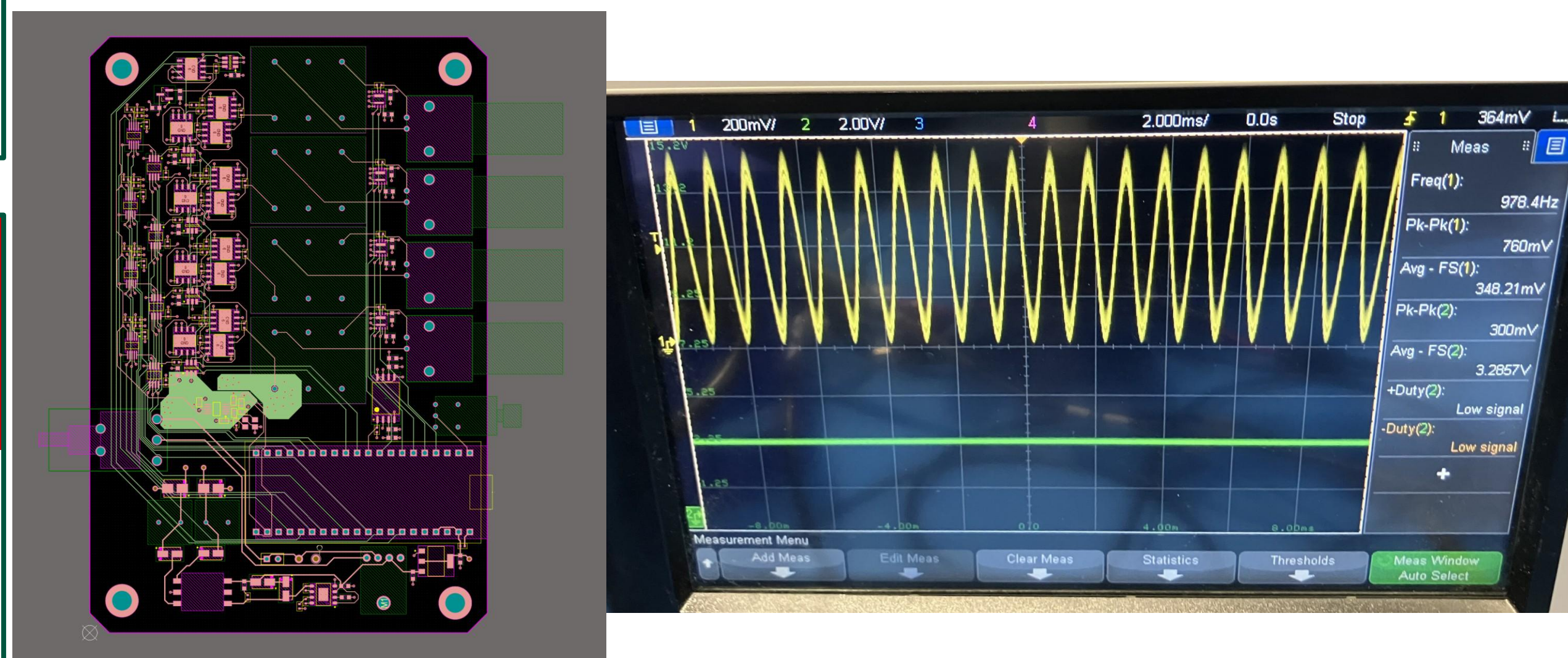
Background and Motivation

Deep brain stimulation (DBS) has emerged as a highly effective neuromodulation technique for treating several neurological disorders, most notably Parkinson's disease, where it has significantly improved motor symptoms and quality of life for thousands of patients. Building on its clinical success, recent research has explored DBS as a potential therapy for severe opioid addiction, with early studies demonstrating promising outcomes in reducing cravings and supporting long-term recovery [1].

The Problem

- Existing stimulators are expensive and not portable
- Participants in studies need to be blinded to whether they are being stimulated
- Current research relies on brand proprietary components

We addressed these issues by making a portable, wireless brain stimulator that can be programmed remotely over Bluetooth and hides stimulation status from the user.



Results

What went right?

- DDS programmability verified
- Isolated ADC feedback verified
- Battery voltage and charging verified
- Bluetooth functionality and web app verified

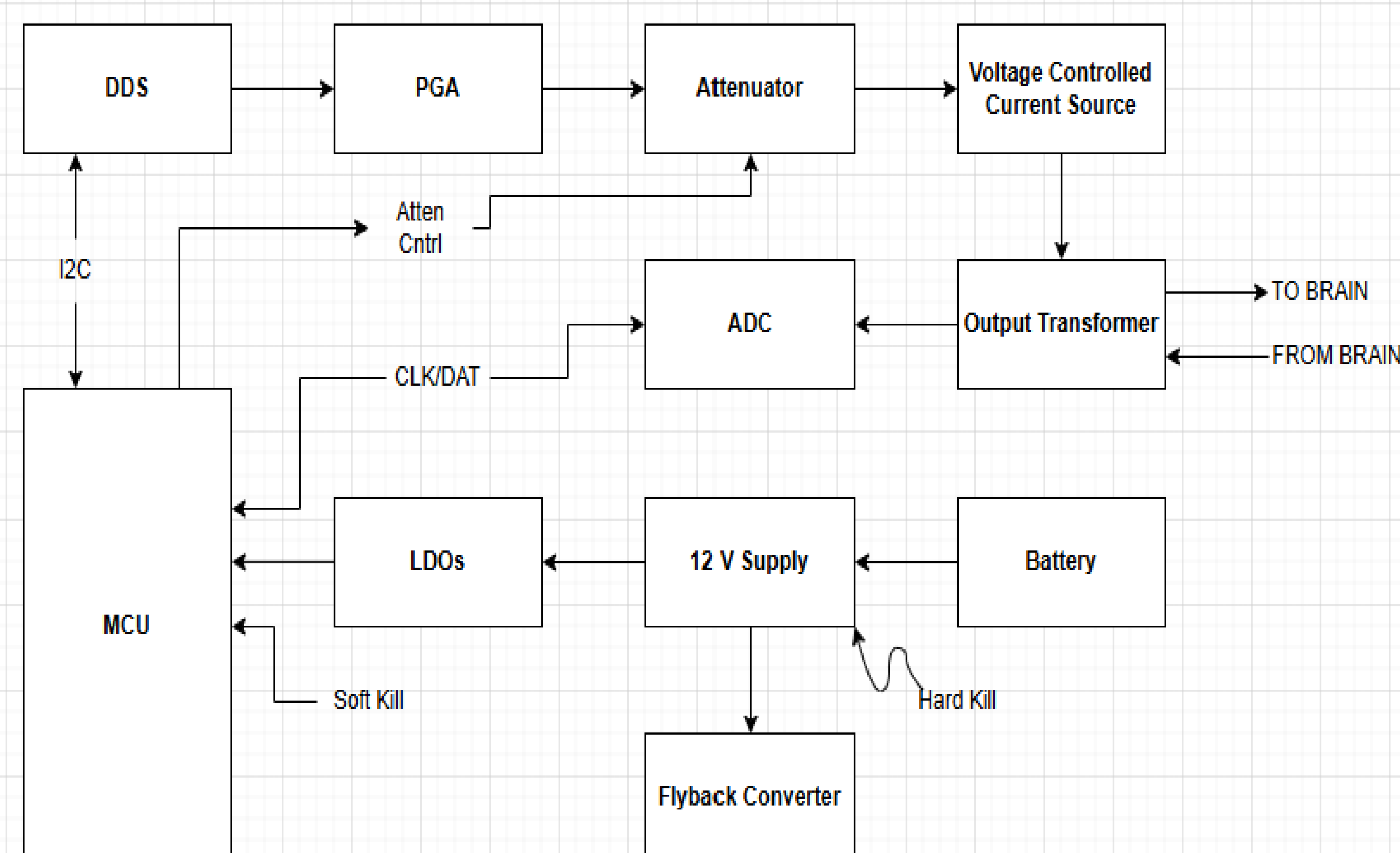
What caused issues?

- Large current draw from +/-30 V boost converter
- Insufficient current sourcing from 12 V boost converter
- Isolated thermal concerns at larger currents

Summary:

We were unable to achieve full functionality, but are hopeful that tweaks to power supply circuitry will make the device function properly

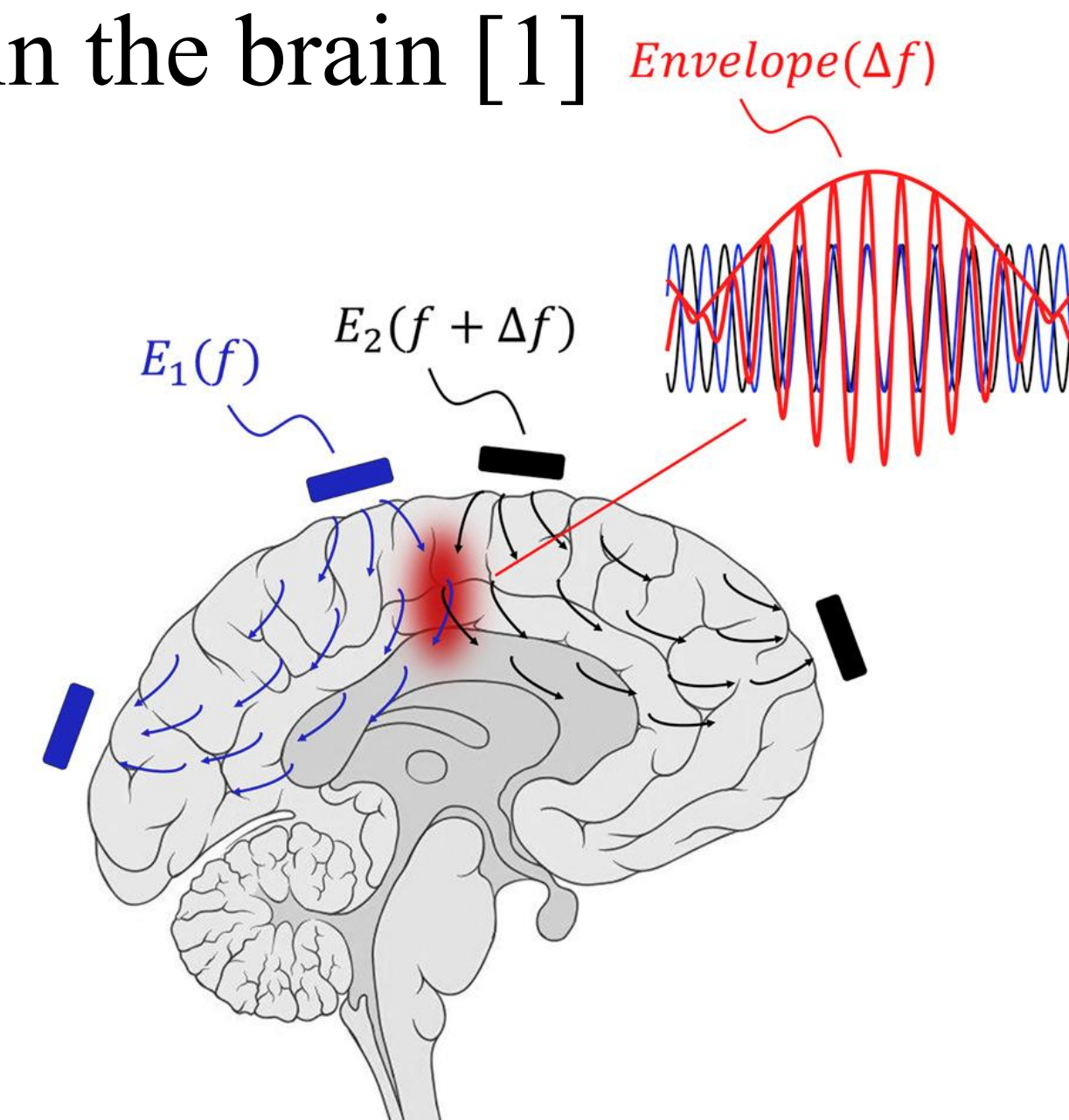
Our Approach



- Pipelined amplifier stages
- Precise control using DDS
- Ramp function to reduce pain
- Transformer isolated four port output
- Onboard battery
- Feedback ADCs for current monitoring

Temporal Interface

- Two slightly different frequencies create a “beat frequency” at the difference of their frequencies
- Using isolated loops and electrodes, can send higher frequency current oscillations along the surface of the head to create low frequency pulses deep in the brain [1]



Next Steps

- Redo power circuitry
- Make custom 12 V converter
- Reduce size of +/- 30 V converter to draw less current
- Update housing to hold battery charger
- Add indicators to device

References

- [1] N. Grossman, et. al. “Noninvasive deep brain stimulation via temporally interfering electric fields,” *Cell*, vol. 169, no. 6, pp 1029-1041, June 1, 2017, doi: 10.1016/j.cell.2017.05.024.

Project done in conjunction with Joshua Brown, Professor of Psychological and Brain Sciences at Indiana University Bloomington